

Woorim Shin

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SUMMARY

Results-driven Computer Science undergraduate (Expected Feb 2027) with a strong passion for defining ambiguous problems and orchestrating the right mix of technologies to build optimal, customer-centric solutions. **Strengths & Motivation:** Driven by a desire to *invent and simplify*, I excel at bridging pedagogical and business planning with scalable AI execution. **Career Plan:** Leveraging my expertise in MLOps, I aspire to grow as a Cloud Data & Analytics Specialist at AWS ProServe—architecting highly refined, cost-effective infrastructure that perfectly answers customer needs.

EDUCATION

Ewha Womans University

B.S. Computer Science and Engineering | **GPA: 3.36/4.5**

Seoul, Korea

Mar. 2023 – Feb. 2027 (Expected)

- **Selected Coursework:** Operating Systems, Computer Architecture, Data Structures, Algorithms, Computer Networks, Database, Machine Learning, Software Engineering

EXPERIENCE

Hiconsy

AI R&D Intern

Seoul, Korea

Jun. 2025 – May 2026

- Led the AI/ML R&D function at a premier Korean CSAT education tech subsidiary, directing research strategy and managing a cross-functional team to deliver AI solutions.
- Architected and deployed a production Korean lecture ASR system end-to-end, and established rigorous QA workflows by building custom annotation tools for non-technical QC staff.

SDIJ Forest Team

Project Leader

Seoul, Korea

Apr. 2023 – May 2025

- Initiated projects focusing on Korean CSAT education planning, expanding scope to technical development as needed to effectively execute product strategies.
- Demonstrated strong technical orchestration by independently developing crawling tools and applying AI technologies to bridge the gap between educational planning and scalable technical execution.
- Engineered a learning analytics automation pipeline (Selenium, Google APIs) replacing manual workflows, slashing data collection time by **80%**, and enabling real-time dashboard monitoring.
- Developed a large-scale NLP pipeline over educational forum data to automate student sentiment analysis and proactive concern detection, directly driving business content strategy.

PROJECTS

Korean CSAT Lecture ASR Customization System

PyTorch, HuggingFace, LoRA, DeepSpeed, Optuna | **Team Size: 1 (Independent Developer)**

Jun. 2025 – Feb. 2026

- Engineered a specialized end-to-end Automatic Speech Recognition (ASR) system to generate highly accurate subtitles specifically for Korean CSAT (College Scholastic Ability Test) lectures at SDIJ.
- Fine-tuned Whisper Large-v3 (1.55B) via LoRA using a custom Korean education tokenizer; authored a distributed training script on 6× RTX 4090 GPUs with DeepSpeed ZeRO-2.
- Independently diagnosed and resolved a severe infinite hallucination bug native to Whisper LoRA fine-tuning by decoupling PAD/EOS tokens, reducing insertion errors by **20.7×** (drafted as a standalone research paper).
- Proactively developed a serverless QA automation layer using Google Apps Script enabling automated SRT parsing, timestamp alignment, and role-based access control, accelerating the data review process.

Energy Cost Model for AI Workloads in Shared Resource Environments

Python, Intel RAPL, nvidia-smi, Linux | **Team Size: 3 (Co-first author)**

Feb. 2026 – Present

- Formulated a resource-aware energy decomposition/attribution model for shared environments, addressing the limitations of traditional allocation-based cloud billing (e.g., AWS EC2).

- Empirically demonstrated that modern AI workloads (YOLO, ResNet, GPT-2) consume **4.7–11× more energy** than traditional workloads (Node.js) despite running under identical resource allocations.
- Constructed a measurement stack utilizing hardware CT sensors (wall power), Intel RAPL, and nvidia-smi to separate baseline and component-specific energy, achieving **<5% error** in multi-tenant approximations.
- **Co-first-authored paper** detailing this attribution model and billing discrepancy.

Educational Q&A Classification System for Korean CSAT

Mar. 2025 – May 2025

OpenAI GPT-4, GPT-3.5 Turbo (fine-tuning), Streamlit | **Team Size: 4 (Project Lead)**

- Led a team of 4 to design a dual-subject cognitive NLP taxonomy from scratch, bridging pedagogical theory with prompt engineering to categorize complex student questions workloads.
- Deployed a production Streamlit dashboard classifying 720+ live questions using fine-tuned GPT-3.5 Turbo and zero-shot GPT-4, accelerating the QA team's response routing.

Korean LLM Benchmark Evaluation

Jan. 2026

Python, OpenAI API, Google Gemini API | **Team Size: 1 (Independent Evaluator)**

- Engineered an automated pipeline (LLM-as-a-judge scoring, regex extraction) to evaluate 6 production LLMs across 4 high-difficulty Korean NLP benchmarks, delivering a corporate report on capability and cost-efficiency.

PUBLICATIONS

Woorim Shin[†], Jiyun Kim, Siyeon Kang, Kyungwoon Cho[†], Hyokyung Bahn. “An Energy Cost Model for AI Workloads under Shared Resource Environments.” *Mathematics*, MDPI (SCIE) – Submitted, Feb 2026. [†]*Equal contribution, co-first author.*

Woorim Shin. “Eliminating Hallucinations in Whisper LoRA Fine-tuning: A Case Study on Korean Lecture Transcription.” *Independent Research (Hiconsy AI R&D – In preparation, 2026)*.

EXTRACURRICULAR & AWARDS

Ewha Startup Contest

Seoul, Korea

3rd Place Winner | *Project: SchedAI (Web-based HR/interview scheduling service)*

Nov. 2025

Google Developer Group on Campus (GDGoC Ewha)

Seoul, Korea

General Member of AI

Sep. 2025 – Present

TECHNICAL SKILLS

Languages: Python, Java, C

AI & Machine Learning: PyTorch, Hugging Face, PEFT/LoRA, DeepSpeed, Whisper, OpenAI/Gemini/Anthropic API

Data & Frameworks: pandas, NumPy, scikit-learn, Streamlit, Selenium, Google Apps Script

Systems & Tools: Linux, Git

Spoken Languages: Korean (Native), English (Advanced)